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7.24

```
MariaDB [(none)]> create database company;  
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [(none)]> use company;  
Database changed
```

```
MariaDB [company]> CREATE TABLE Department (deptNo INT PRIMARY KEY,  
deptName VARCHAR(50) NOT NULL, mgrEmpNo INT);  
Query OK, 0 rows affected (0.035 sec)
```

```
MariaDB [company]> create table Employee(empNo VARCHAR(5) PRIMARY KEY,  
fName varchar(30) not null, lName varchar(30) not null, address  
varchar(100), DOB DATE, sex CHAR(1) check (sex in ('M','F')), position  
varchar(30) check (position in ('Manager', 'Team Leader', 'Analyst',  
'Software Developer')), deptNo varchar(5), FOREIGN KEY (deptNo)  
REFERENCES Department(deptNo));  
Query OK, 0 rows affected (0.029 sec)
```

```
MariaDB [company]> CREATE TABLE Project (projNo VARCHAR(10) PRIMARY  
KEY, projName VARCHAR(50) NOT NULL,  
-> deptNo VARCHAR(5), FOREIGN KEY (deptNo) REFERENCES  
Department(deptNo));  
Query OK, 0 rows affected (0.029 sec)
```

```
MariaDB [company]> CREATE TABLE WorksOn (empNo VARCHAR(5), projNo  
VARCHAR(10), dateWorked DATE, hoursWorked INT CHECK (hoursWorked  
BETWEEN 0 AND 40), PRIMARY KEY(empNo, projNo, dateWorked), FOREIGN  
KEY(empNo) REFERENCES Employee(empNo), FOREIGN KEY(projNo) REFERENCES  
Project(projNo));  
Query OK, 0 rows affected (0.030 sec)
```

```
MariaDB [company]> ALTER TABLE Department ADD CONSTRAINT FK_Manager  
FOREIGN KEY (mgrEmpNo) REFERENCES Employee(empNo);  
Query OK, 0 rows affected (0.054 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [company]> INSERT INTO Department VALUES ('D1','Human  
Resources',NULL), ('D2','Information Technology',NULL),  
('D3','Finance',NULL), ('D4','Marketing',NULL);  
Query OK, 4 rows affected (0.001 sec)  
Records: 4 Duplicates: 0 Warnings: 0
```

```
MariaDB [company]> select * from Department;
```

deptNo	deptName	mgrEmpNo
D1	Human Resources	NULL
D2	Information Technology	NULL
D3	Finance	NULL
D4	Marketing	NULL

4 rows in set (0.000 sec)

```
MariaDB [company]> INSERT INTO Employee
-> VALUES
-> ('E1','John','Mensah','Accra','1995-06-10','M','Manager','D2'),
->
-> ('E2','Mary','Owusu','Tema','1998-08-15','F','Software
Developer','D2'),
->
->
-> ('E3','Daniel','Boateng','Kumasi','1997-03-20','M','Manager','D3'),
->
->
-> ('E4','Sarah','Asante','Takoradi','1999-01-25','F','Manager','D1'),
->
-> ('E5','Michael','Aidoo','Cape Coast','2000-11-30','M','Software
Developer','D2'),
->
-> ('E6','Linda','Kusi','Accra','1996-04-17','F','Manager','D4');
Query OK, 6 rows affected (0.002 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

```
MariaDB [company]> select * from Employee;
```

empNo	fName	lName	address	DOB	sex	position
E1	John	Mensah	Accra	1995-06-10	M	Manager
D2						
E2	Mary	Owusu	Tema	1998-08-15	F	Software Developer
D2						
E3	Daniel	Boateng	Kumasi	1997-03-20	M	Manager
D3						
E4	Sarah	Asante	Takoradi	1999-01-25	F	Manager
D1						
E5	Michael	Aidoo	Cape Coast	2000-11-30	M	Software Developer
D2						
E6	Linda	Kusi	Accra	1996-04-17	F	Manager
D4						

```
+-----+-----+
6 rows in set (0.000 sec)
```

```
MariaDB [company]> UPDATE Department SET mgrEmpNo='E4' WHERE
deptNo='D1';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [company]> UPDATE Department SET mgrEmpNo='E1' WHERE
deptNo='D2';
Query OK, 1 row affected (0.000 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [company]> UPDATE Department SET mgrEmpNo='E3' WHERE
deptNo='D3';
Query OK, 1 row affected (0.000 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [company]> UPDATE Department SET mgrEmpNo='E6' WHERE
deptNo='D4';
Query OK, 1 row affected (0.001 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [company]> select * from Department;
```

```
+-----+-----+
| deptNo | deptName          | mgrEmpNo |
+-----+-----+
| D1     | Human Resources   | E4       |
| D2     | Information Technology | E1       |
| D3     | Finance          | E3       |
| D4     | Marketing         | E6       |
+-----+-----+
```

4 rows in set (0.000 sec)

```
MariaDB [company]> INSERT INTO Project VALUES ('SCCS','Student Course
System','D2'), ('IMS','Inventory Management System','D3'),
('HRMS','Human Resource System','D1'), ('MKT','Marketing
Website','D4'), ('APP','Mobile Application','D2');
Query OK, 5 rows affected (0.003 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
MariaDB [company]> select * from Project;
```

```
+-----+-----+
| projNo | projName          | deptNo |
+-----+-----+
| APP    | Mobile Application | D2     |
| HRMS   | Human Resource System | D1     |
| IMS    | Inventory Management System | D3     |
| MKT    | Marketing Website  | D4     |
| SCCS   | Student Course System | D2     |
+-----+-----+
```

```
+-----+-----+-----+
5 rows in set (0.000 sec)
```

```
MariaDB [company]> INSERT INTO WorksOn VALUES
('E1','SCCS','2026-06-01',8), ('E2','SCCS','2026-06-01',7),
('E2','APP','2026-06-02',6), ('E3','IMS','2026-06-03',8),
('E4','HRMS','2026-06-02',7), ('E5','SCCS','2026-06-03',9),
('E5','APP','2026-06-04',8), ('E6','MKT','2026-06-05',6),
('E1','APP','2026-06-06',5), ('E6','MKT','2026-06-07',8);
Query OK, 10 rows affected (0.001 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
MariaDB [company]> select * from WorksOn;
```

empNo	projNo	dateWorked	hoursWorked
E1	APP	2026-06-06	5
E1	SCCS	2026-06-01	8
E2	APP	2026-06-02	6
E2	SCCS	2026-06-01	7
E3	IMS	2026-06-03	8
E4	HRMS	2026-06-02	7
E5	APP	2026-06-04	8
E5	SCCS	2026-06-03	9
E6	MKT	2026-06-05	6
E6	MKT	2026-06-07	8

```
10 rows in set (0.000 sec)
```

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```
MariaDB [company]> CREATE VIEW FemaleManagerProjects AS SELECT
p.projNo, p.projName, d.deptName, e.empNo,
-> e.fName, e.lName FROM Project p JOIN Department d ON
p.deptNo=d.deptNo JOIN Employee e
-> ON d.mgrEmpNo=e.empNo WHERE e.sex='F' ORDER BY p.projNo;
Query OK, 0 rows affected (0.019 sec)
```

```
MariaDB [company]> select * from FemaleManagerProjects;
```

projNo	projName	deptName	empNo	fName	lName
HRMS	Human Resource System	Human Resources	E4	Sarah	Asante
MKT	Marketing Website	Marketing	E6	Linda	Kusi

```

+-----+-----+-----+-----+-----+
+-----+
2 rows in set (0.002 sec)

```

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```

MariaDB [company]> CREATE VIEW EmployeeProjectHours AS SELECT e.empNo,
e.fName, e.lName, p.projName, w.hoursWorked FROM Employee e JOIN
WorksOn w ON e.empNo=w.empNo JOIN Project p ON w.projNo=p.projNo;
Query OK, 0 rows affected (0.022 sec)

```

```

MariaDB [company]> select * from EmployeeProjectHours;

```

```

+-----+-----+-----+-----+-----+
+-----+
| empNo | fName  | lName  | projName                |
hoursWorked |
+-----+-----+-----+-----+-----+
+-----+
| E1    | John   | Mensah | Mobile Application      |
5 |
| E1    | John   | Mensah | Student Course System   |
8 |
| E2    | Mary   | Owusu  | Mobile Application      |
6 |
| E2    | Mary   | Owusu  | Student Course System   |
7 |
| E3    | Daniel | Boateng | Inventory Management System |
8 |
| E4    | Sarah  | Asante | Human Resource System    |
7 |
| E5    | Michael | Aidoo  | Mobile Application      |
8 |
| E5    | Michael | Aidoo  | Student Course System    |
9 |
| E6    | Linda  | Kusi   | Marketing Website       |
6 |
| E6    | Linda  | Kusi   | Marketing Website       |
8 |
+-----+-----+-----+-----+-----+
+-----+
10 rows in set (0.001 sec)

```

7.27

```

MariaDB [company]> create VIEW EmpProject(empNo, projNo, totalHours)
AS SELECT w.empNo, w.projNo, SUM(hoursWorked) FROM Employee e, Project
p, WorksOn w WHERE e.empNo = w.empNo AND p.projNo = w.projNo GROUP BY
w.empNo, w.projNo;
Query OK, 0 rows affected (0.024 sec)

```

MariaDB [company]> select * from EmpProject;

empNo	projNo	totalHours
E1	APP	5
E1	SCCS	8
E2	APP	6
E2	SCCS	7
E3	IMS	8
E4	HRMS	7
E5	APP	8
E5	SCCS	9
E6	MKT	14

9 rows in set (0.001 sec)

MariaDB [company]> select projNo FROM EmpProject WHERE projNo = 'SCCS';

projNo
SCCS
SCCS
SCCS

3 rows in set (0.003 sec)

MariaDB [company]> select COUNT(projNo) from EmpProject WHERE empNo = 'E1';

COUNT(projNo)
2

1 row in set (0.001 sec)

MariaDB [company]> select empNo, totalHours FROM EmpProject GROUP BY empNo;

empNo	totalHours
E1	5
E2	6
E3	8
E4	7
E5	8
E6	14

6 rows in set (0.001 sec)